Aim :- NTP Server Configuration using Chrony (RHEL 9)

Synchronize the system date and time with a central server using NTP service.

1) Manual time and date check and change :

```
[root@ntpserver ~]# date
Thu May 14 11:42:40 PM IST 2026
[root@ntpserver ~]# date -s "15 May 2026 15:30:00"
Fri May 15 03:30:00 PM IST 2026
[root@ntpserver ~]#
[root@ntpserver ~]#
```

2)permanent time set:

Step 1 :- Which timezone are we in?

```
[root@ntpserver ~]# timedatectl
               Local time: Fri 2026-05-15 15:31:08 IST
           Universal time: Fri 2026-05-15 10:01:08 UTC
                 RTC time: Thu 2026-05-14 18:15:06
                Time zone: Asia/Kolkata (IST, +0530)
System clock synchronized: no
              NTP service: active
          RTC in local TZ: no
[root@ntpserver ~]#
[root@ntpserver ~]#
```

Step 2 :-My server is in Germany for time set i follow process :-

```
[root@ntpserver ~]#
[root@ntpserver ~]# tzselect
Please identify a location so that time zone rules can be set correctly.
Please select a continent, ocean, "coord", or "TZ".
 1) Africa
 2) Americas
 3) Antarctica
 4) Asia
 5) Atlantic Ocean
 6) Australia
 7) Europe
 8) Indian Ocean
 9) Pacific Ocean
10) coord - I want to use geographical coordinates.
11) TZ - I want to specify the timezone using the Posix TZ format.
#? 7
Please select a country whose clocks agree with yours.
 1) Albania              11) Denmark              21) Isle of Man          31) Montenegro           41) Slovakia
 2) Andorra              12) Estonia              22) Italy                32) Netherlands          42) Slovenia
 3) Austria              13) Finland              23) Jersey               33) North Macedonia      43) Spain
 4) Belarus              14) France               24) Latvia               34) Norway               44) Svalbard & Jan Mayen
 5) Belgium              15) Germany              25) Liechtenstein        35) Poland               45) Sweden
 6) Bosnia & Herzegovina 16) Gibraltar            26) Lithuania            36) Portugal             46) Switzerland
 7) Britain (UK)         17) Greece               27) Luxembourg           37) Romania              47) Turkey
 8) Bulgaria             18) Guernsey             28) Malta                38) Russia               48) Ukraine
 9) Croatia              19) Hungary              29) Moldova              39) San Marino           49) Vatican City
10) Czech Republic       20) Ireland              30) Monaco               40) Serbia               50) Åland Islands
#? 15
Please select one of the following timezones.
1) Büsingen
2) most of Germany
#? 2

The following information has been given:

        Germany
        most of Germany

Therefore TZ='Europe/Berlin' will be used.
Selected time is now:   Fri May 15 12:06:57 CEST 2026.
Universal Time is now:  Fri May 15 10:06:57 UTC 2026.
Is the above information OK?
1) Yes
2) No
#? 1

You can make this change permanent for yourself by appending the line
        TZ='Europe/Berlin'; export TZ
to the file '.profile' in your home directory; then log out and log in again.

Here is that TZ value again, this time on standard output so that you
can use the /usr/bin/tzselect command in shell scripts:
Europe/Berlin
[root@ntpserver ~]#
[root@ntpserver ~]#
```

Step 3 :-Now I change time and verify :-

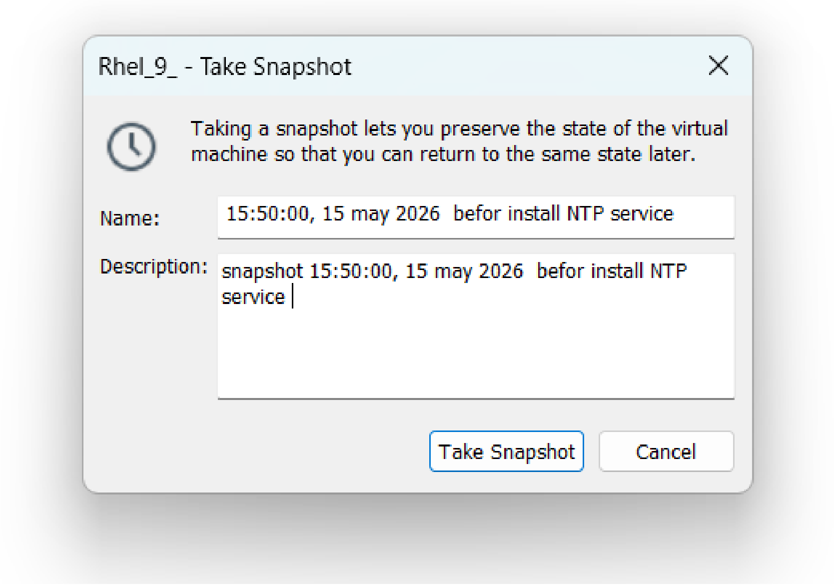
```
[root@ntpserver ~]# timedatectl set-timezone Europe/Berlin
[root@ntpserver ~]#
[root@ntpserver ~]#
[root@ntpserver ~]#
[root@ntpserver ~]#
[root@ntpserver ~]# timedatectl
               Local time: Fri 2026-05-15 12:43:58 CEST
           Universal time: Fri 2026-05-15 10:43:58 UTC
                 RTC time: Thu 2026-05-14 18:57:55
                Time zone: Europe/Berlin (CEST, +0200)
System clock synchronized: no
              NTP service: active
          RTC in local TZ: no
[root@ntpserver ~]#
```

(In a production environment, typically one central NTP server is used, and all systems synchronize their time with that server. To achive this we install, configure and verify the NTP service . the package name which install for synchronization “chrony”)

For synchronization NTP service install ,configure and verify

Step 1: log in

Step 2: Take backup and restore plan



Step 3:check Network and configure

```
[root@ntpserver ~]# hostname
ntpserver
[root@ntpserver ~]# nmtui
[root@ntpserver ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.100  netmask 255.255.255.0  broadcast 192.168.0.255
        inet6 fe80::20c:29ff:fe36:8472  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:36:84:72  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 82  bytes 9017 (8.8 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 174  bytes 14219 (13.8 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 174  bytes 14219 (13.8 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Step 4: configure required Repository :-

```
[root@ntpserver ~]# yum search chrony
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 0:02:00 ago on Fri 15 May 2026 01:17:49 PM CEST.
==================================================================================== Name Exactly Matched: chrony ==============================================
chrony.x86_64 : An NTP client/server
[root@ntpserver ~]# yum install -y chrony.x86_64
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 0:02:22 ago on Fri 15 May 2026 01:17:49 PM CEST.
Package chrony-4.6.1-2.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ntpserver ~]#
```

Step 5: take backup of configure file :-

```
[root@ntpserver ~]# vi /etc/chrony.conf
[root@ntpserver ~]# cp /etc/chrony.conf /tmp/chrony.bkp
[root@ntpserver ~]#
[root@ntpserver ~]#
```

Step 6: edit configuration file :-

```
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (https://www.pool.ntp.org/join.html).
#pool 2.rhel.pool.ntp.org iburst
server my_ntp_srv.radical.com  iburst

# Use NTP servers from DHCP.
sourcedir /run/chrony-dhcp

# Record the rate at which the system clock gains/losses time.
driftfile /var/lib/chrony/drift

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
rtcsync

# Enable hardware timestamping on all interfaces that support it.
#hwtimestamp *

# Increase the minimum number of selectable sources required to adjust
# the system clock.
#minsources 2

# Allow NTP client access from local network.
```

Step 7: start and enable service :-

```
[root@ntpserver ~]# systemctl start chronyd
[root@ntpserver ~]# systemctl enable chronyd
[root@ntpserver ~]# systemctl restart chronyd
[root@ntpserver ~]#
```

Step 8:check status :-

```
[root@ntpserver ~]#
[root@ntpserver ~]# systemctl status chronyd
● chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-05-15 03:47:16 CEST; 25s ago
       Docs: man:chronyd(8)
             man:chrony.conf(5)
    Process: 3145 ExecStart=/usr/sbin/chronyd $OPTIONS (code=exited, status=0/SUCCESS)
   Main PID: 3147 (chronyd)
      Tasks: 1 (limit: 10312)
     Memory: 1.0M (peak: 1.8M)
        CPU: 42ms
     CGroup: /system.slice/chronyd.service
             └─3147 /usr/sbin/chronyd -F 2

May 15 03:47:16 ntpserver systemd[1]: Starting NTP client/server...
May 15 03:47:16 ntpserver chronyd[3147]: chronyd version 4.6.1 starting (+CMDMON +NTP +REFCLOCK +RTC +PRIVDROP +SCFILTER +SIGND +ASYNCDNS +NTS +SECHASH +IPV6 +DE
May 15 03:47:16 ntpserver chronyd[3147]: Loaded 0 symmetric keys
May 15 03:47:16 ntpserver chronyd[3147]: Using right/UTC timezone to obtain leap second data
May 15 03:47:16 ntpserver chronyd[3147]: Frequency -11.742 +/- 55.528 ppm read from /var/lib/chrony/drift
May 15 03:47:16 ntpserver chronyd[3147]: Loaded seccomp filter (level 2)
May 15 03:47:16 ntpserver systemd[1]: Started NTP client/server.
lines 1-20/20 (END)
```

Step 9: Start daemon and make it Persistent :-

```
[root@ntpserver ~]# systemctl enable --now chronyd.service
[root@ntpserver ~]#
```

Step 10: Verify

```
[root@ntpserver ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
[root@ntpserver ~]#
[root@ntpserver ~]#
[root@ntpserver ~]#
```

(synchronization did not start because the service was not start, once the correct service name used, the service will start and synchronization will begin.)